Pricing AT1 Contingent Convertibles as a Claim on Bank Capital

Model, numerical method, and a calibration to BNP Paribas's USD AT1s

Paolo Mastrogiacomo
`Credit-Quant-Research / at1-coco-pricer`

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Abstract

An Additional Tier 1 (AT1) bond is a perpetual, deeply subordinated bank bond whose cash flows depend on the issuer's capital: coupons are cut when capital enters the regulatory buffer, principal is converted or written down at a capital trigger or at the regulator's discretion, and the issuer calls only when refinancing is cheaper. We take the Common Equity Tier 1 (CET1) ratio as the state variable, add a mean-reverting issuer spread that drives both the call decision and the regulatory write-down (PONV) hazard, and price by Monte Carlo with the PONV hazard integrated analytically along each path. Calibrated to BNP Paribas's three USD AT1s (September 2026), with the FRED Treasury curve and 50 quarters of CET1 history, one implied hazard of about 2.3% a year prices all three bonds. About 240bp of the $\approx$ 270–280bp spread is compensation for PONV, tail and liquidity risk; mechanical conversion explains $\approx$ 4bp. The price is flat in capital above the MDA threshold ($\approx$ 0.45 points per point of CET1) and steep around it ($\approx$ 4 points). A bootstrap shows that the speed of capital mean reversion is the weakest input: the half-life of a capital shock is anywhere between 1.5 and 11 years, and the capital stress results scale by a factor of six across that range. The bond cross-section is consistent only with the faster end, which sets the base case.

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1 The instrument and the idea

1.1 What an AT1 is

An AT1 bond counts as going-concern regulatory capital for a bank. Its terms make it behave partly like equity:

1. **Perpetual, with issuer calls.** There is no maturity. The issuer may redeem at par on call dates; for the bonds studied here, only on *reset dates*, every five years from the first call. If not called, the coupon resets to a reference rate (5-year US Treasury, CMT) plus the margin fixed at issue. Investors are exposed to *extension*: the issuer calls when it is cheap to refinance, and does not when it is expensive, which is exactly when investors would want their money back.

2. **Discretionary, non-cumulative coupons.** Coupons may be cancelled at any time and are lost for good. They are also mechanically restricted when CET1 falls inside the *combined buffer requirement*: the Maximum Distributable Amount (MDA) rule of CRD IV.

3. **Loss absorption.** If CET1 falls below a contractual trigger (5.125% here), the bond converts into shares (or is written down). The resolution authority can also impose losses at the *Point of Non-Viability* (PONV) while CET1 is still above the trigger: Credit Suisse's AT1s were written to zero in March 2023 with a reported CET1 ratio of about 14%.

1.2 Why capital is the state variable

Trigger, coupon restriction and (partly) the call decision are all written on the CET1 ratio. Modelling CET1 directly makes these three risks consistent by construction: the same simulated path that approaches the MDA threshold also cuts coupons, discourages a call, and raises the chance of loss absorption. The alternative, an equity-barrier model, needs a mapping from share price to capital that is unobservable and unstable.

Capital alone is not enough, however. The market prices AT1s far below what historical capital volatility would justify (§7), and the call decision depends on market spreads, not only on capital. We therefore add a second state variable, the issuer's AT1 spread, which (i) drives the call decision and (ii) scales the PONV hazard.

2 State dynamics

2.1 CET1 ratio

Let C_t be the CET1 ratio in percentage points. We use a mean-reverting jump-diffusion with downward jumps:

$$dC_t = \kappa(\theta - C_t)\,dt + \sigma\,dW_t - J\,dN_t, \tag{1}$$

where N_t is a Poisson process with intensity λ_J (stress events) and jump sizes are i.i.d.

$$J = m_J \exp\!\big(\sigma_J Z - \tfrac{1}{2}\sigma_J^2\big), \qquad Z \sim N(0,1), \qquad \mathbb{E}[J] = m_J.$$

Mean reversion to θ represents the bank's capital management (retained earnings, distribution policy, management of risk-weighted assets); the target θ is management's stated CET1 objective. Jumps are one-sided because losses arrive suddenly while recapitalisation is gradual.

The *mechanical trigger time* is $\tau_m = \inf\{t : \ C_t \le c_{\text{trig}}\}$.

2.2 Issuer AT1 spread

Let S_t be the market level of the issuer's new-issue AT1 spread (decimal). It follows an Ornstein–Uhlenbeck process correlated with capital shocks,

$$dS_t = a(\bar{s} - S_t)\,dt + \eta\,dB_t, \qquad d\langle B, W\rangle_t = \rho\,dt, \quad \rho < 0, \tag{2}$$

and the issuer's *effective* spread adds a capital premium once CET1 is inside the buffer:

$$s_t = S_t + \beta\,\big(\bar{c}_{\text{MDA}} - C_t\big)^+, \tag{3}$$

where $\bar{c}_{\text{MDA}}$ is the MDA threshold defined below. The effective spread s_t is the rate at which the bank could issue a new AT1 at time t.

3 Contract features as functionals of the state

3.1 MDA coupon restriction

Let b be the MDA floor (Pillar 1 plus the CET1 part of Pillar 2R) and w the combined buffer (conservation, systemic and countercyclical buffers), so that $\bar{c}_{\text{MDA}} = b + w$. With $q = (C - b)/w$

the position inside the buffer, the fraction of the scheduled coupon that can be paid is

$$\phi(C) = \begin{cases} 1 & C \geq b + w, \\ 0.6 & q \in [0.75, 1), \\ 0.4 & q \in [0.50, 0.75), \\ 0.2 & q \in [0.25, 0.50), \\ 0 & q < 0.25 \text{ or } C < b. \end{cases} \tag{4}$$

This is a stylised version of the quartile schedule; it makes coupon-cancellation risk emerge from the same paths as trigger risk.

3.2 Call and extension

Let $T_1 < T_2 < \dots$ be the reset (call) dates and m the reset margin over the 5-year reference rate. The issuer calls at the first reset date T_k at which

$$m > s_{T_k} \qquad \text{and} \qquad C_{T_k} \geq \bar{c}_{\text{MDA}} \qquad \text{and} \qquad T_k < \tau_m, \tag{5}$$

that is, when keeping the bond costs more than issuing a new one, provided the regulator would approve the call (CET1 above the MDA threshold) and the bond is still alive. Denote the call time by τ_c ($+\infty$ if never). If $\eta > 0$ the call is random even for a healthy bank; the low-margin bond is the first to be extended when spreads widen.

3.3 Conversion

At τ_m the bond converts into shares at the higher of the market price and a floor price P_{floor}. The investor receives shares worth $F \cdot S^{\text{eq}}_{\tau_m}/P_{\text{floor}}$ (when the share trades below the floor, as it will at the trigger). The model uses a stressed share price $S^{\text{eq}}_{\tau_m} = \xi\, S^{\text{eq}}_0 X_0$ (converted to USD at X_0), so the recovery is

$$R = \min\left(1,\ \frac{\xi\, S^{\text{eq}}_0 X_0}{P_{\text{floor}}}\right). \tag{6}$$

3.4 PONV: a hazard linked to the spread

The regulator may impose losses before the trigger. We model this as a doubly-stochastic (Cox) event with intensity proportional to the effective spread,

$$\lambda_t = \lambda\, \frac{s_t}{s_0}, \tag{7}$$

so λ is today's hazard, and the hazard rises when spreads widen or capital falls inside the buffer. Recovery at PONV is zero. Linking the hazard to s_t is what makes extension consistent: the states in which the issuer extends ($s_t > m$) are exactly those in which the bond is riskier and worth less. With a constant hazard, extension would be valued at a discount rate that does not reflect the state, and could even look valuable to the investor.

4 Valuation

4.1 Pricing equation

Let r be the flat, continuously compounded risk-free rate, F the face and t_i the coupon dates, paid at frequency f. The scheduled coupon is $c(t_i) = c_0$ up to and including T_1 and $c(t_i) = y^{\text{ref}} + m$

after it. Let τ_P be the PONV time. The dirty price is

$$V_0 = \mathbb{E}\Bigg[\sum_{t_i} \mathbf{1}_{\{t_i<\tau_m,\ t_i\le\tau_c,\ t_i<\tau_P\}} \frac{\phi(C_{t_i})\, c(t_i)\, F}{f}\, e^{-rt_i} + \mathbf{1}_{\{\tau_c<\tau_m,\ \tau_c<\tau_P\}}\, F\, e^{-r\tau_c} + \mathbf{1}_{\{\tau_m<\tau_c,\ \tau_m<\tau_P\}}\, R\, F\, e^{-r\tau_m} \Bigg] + \text{tail}. \tag{8}$$

Paths alive at the horizon $H = 40$ years receive the value of a perpetuity at the reset coupon, $F(y^{\text{ref}} + m)/(r + \lambda_H)$, discounted to today.

4.2 Conditional Monte Carlo for the PONV hazard

Proposition 1. *Let $\mathcal{G}$ be the σ-algebra generated by the paths of (C, S). Conditional on $\mathcal{G}$, the PONV time has intensity (7), so*

$$\Pr(\tau_P > t \mid \mathcal{G}) = \exp\Big(-\lambda \int_0^t \frac{s_u}{s_0}\, du\Big) \equiv e^{-\lambda I_t}.$$

Hence

$$V_0(\lambda) = \mathbb{E}\Bigg[\sum_{t_i} \text{CF}_i\, e^{-rt_i - \lambda I_{t_i}} + \text{CF}_{\text{end}}\, e^{-rT_{\text{end}} - \lambda I_{T_{\text{end}}}} \Bigg], \tag{9}$$

where CF_i are the coupon cash flows and CF_{end} is the call, conversion or tail payment at the end of each path's life, all computed without reference to τ_P.

Proof. Condition each indicator $\mathbf{1}_{\{t<\tau_P\}}$ in (8) on $\mathcal{G}$ (tower property); every other term is $\mathcal{G}$-measurable. □

Consequences for the numerics:

- PONV is never simulated as an event, which removes its sampling noise (Rao–Blackwellisation).
- On a fixed set of paths, $V_0(\lambda)$ is smooth and strictly decreasing in λ (because $I_t > 0$). The market-implied hazard is therefore the unique root of $V_0(\lambda) = P^{\text{clean}} + \text{AI}$ and is found with Brent's method.
- Cash flows and I_t are stored per path once; each evaluation of $V_0(\lambda)$ is a vectorised sum.

4.3 Discretisation

The grid is monthly ($\Delta = 1/12$) over 40 years, with 20,000 paths. CET1 uses an Euler step with a Bernoulli jump of probability $1 - e^{-\lambda_J \Delta}$; the spread uses the exact OU transition

$$S_{n+1} = \bar{s} + (S_n - \bar{s})e^{-a\Delta} + \eta\sqrt{\tfrac{1-e^{-2a\Delta}}{2a}}\,\big(\rho Z_n^C + \sqrt{1-\rho^2}\, Z_n^{\perp}\big).$$

Coupons are discounted at their exact dates; path events (trigger, call) are read on the grid. Market prices are clean, so the model value is compared with the clean price plus 30/360 accrued interest (the accrued amount matches Capital IQ to three decimals).

4.4 Common random numbers

All primitive random numbers are drawn once and reused across scenarios, bumps and feature switches. Differences between two prices (Greeks, stress P&L, spread components) therefore carry no Monte Carlo noise of their own.

5 Calibration

5.1 Capital dynamics from history

The CET1 series is quarterly from 2014Q1 to 2026Q2 (50 observations). Earlier data follow Basel II/2.5 definitions and are not comparable. The 2023Q1 change (+1.2pp, sale of Bank of the West) is a disposal, not a capital shock, and that transition is excluded from the likelihood. The OU transition density is Gaussian for any step Δ_k,

$$C_{k+1} \mid C_k \sim N\Big(\theta + (C_k - \theta)e^{-\kappa\Delta_k},\ \tfrac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa\Delta_k}\big)\Big),$$

so the likelihood is exact. Fixing $\theta = 13\%$ (BNP's target) gives $\hat{\kappa} = 0.165$ and $\hat{\sigma} = 0.44\text{pp}/\sqrt{\text{yr}}$; κ is weakly identified and the base case uses $\kappa = 0.35$ (§7.6), with $\sigma = 0.47$ re-estimated given κ. The 2014Q2 fall of 0.76pp (US sanctions settlement) is an example of the stress events the jumps represent. The jumps cannot be identified from this sample; we use $\lambda_J = 0.12$, $m_J = 2.2$pp, $\sigma_J = 0.5$, whose 95th-percentile jump (4.4pp) matches BNP's CET1 depletion in the EBA 2025 adverse scenario (4.2pp, fully loaded).

5.2 Regulatory geometry

From the 2025 requirement stack, $b = 4.5 + 1.14 = 5.64\%$ and $w = 2.5 + 1.5 + 0.73 + 0.14 = 4.87\%$ (conservation, G-SIB, countercyclical and systemic-risk buffers), so $\bar{c}_{\text{MDA}} = 10.51\%$. The trigger is $c_{\text{trig}} = 5.125\%$ of group CET1.

5.3 Market inputs

The inputs are:

- the clean price of each bond (Capital IQ evaluated prices);
- the FRED constant-maturity Treasury curve (DGS5, DGS7, DGS10): the 5-year CMT is the reset reference, and each bond is discounted at the yield interpolated at its first call date;
- the AT1 spread S_0, the average of (bond yield − call-matched Treasury);
- the latest published CET1 ratio, with a 35-day publication lag;
- the share price.

The spread process uses $a = 0.5$, $\bar{s}$ equal to the sample mean (275bp), $\eta = 100$bp and $\rho = -0.3$; the capital premium is $\beta = 1\%$ per point of CET1 and the share stress is $\xi = 0.2$.

5.4 Implied hazard and cross-sectional identification

For each date and bond j we solve $V_0^{(j)}(\lambda_j) = P_j$. The three bonds share issuer, capital and trigger, so a correct model should give $\lambda_1 = \lambda_2 = \lambda_3$. The dispersion $D = \max_j \lambda_j - \min_j \lambda_j$ is

a specification test, and it can be used to ask which values of the other parameters (the spread volatility η, the capital mean reversion κ) the three prices are consistent with.

6 Analytics

6.1 Spread decomposition

Switch the features on one at a time and convert each price into a yield to the first call date, $y^{(k)}$ (semi-annual, bullet to T_1):

k	price $V^{(k)}$
0	certain call at T_1, full coupons, no conversion, $\lambda = 0$ (risk-free)
1	+ PONV hazard λ
2	+ MDA coupon restriction
3	+ conversion at the trigger
4	+ economic call rule (5) = market price

Because $V^{(0)}$ is discounted at r, $y^{(0)}$ is the Treasury yield, and the increments $y^{(k)} - y^{(k-1)}$ telescope to the market yield-to-call spread. The order is chosen so that every feature is valued with risky discounting and extension comes last, with all other features in place.

6.2 Call probability and the cost of the call option

$\Pr(\tau_c = T_1)$ is read from the paths. The cost of the issuer's option is $V^{\text{call at } T_1} - V$. Negative convexity is measured as the extra loss for a +50bp shift in the spread level (with the hazard scaled accordingly), relative to a bond whose call at T_1 is certain.

6.3 Stress and capital delta

Each scenario shifts C_0, the spread (S_0 and $\bar{s}$, with λ scaled by S) or the Treasury level, and reprices on the same paths. The capital delta $\partial V / \partial C_0$ is a finite difference on a grid of C_0.

7 Results: BNP Paribas, 25 September 2026

7.1 Capital

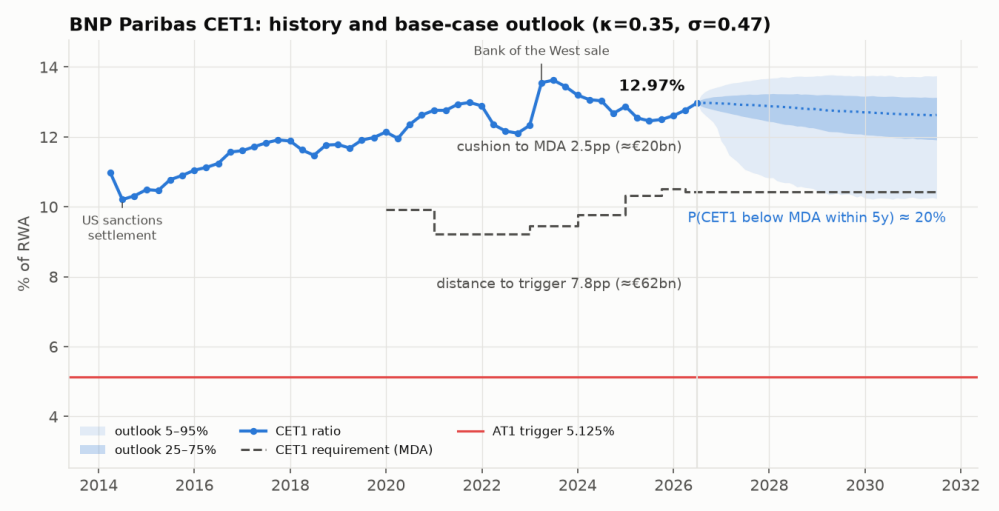


Figure 1: CET1 ratio since 2014, CET1 requirement (MDA threshold), AT1 trigger and the base-case 5-year outlook (median, 50% and 90% bands of the simulated CET1).

CET1 was 12.97% at June 2026 against a 10.43% requirement: a cushion of 2.5pp ($\approx$ EUR 20bn). The trigger is 7.8pp lower ($\approx$ EUR 62bn of losses). Under the base-case dynamics, about 20% of the simulated paths fall below the MDA threshold at some point in the next five years, while the trigger is almost never reached by diffusion alone: capital risk for these bonds is first a coupon risk, then a conversion risk.

7.2 The three bonds and the implied hazard

	6.875% NC33	7.20% NC36	7.45% NC35
Reset margin (bp)	285	294	313
Clean price	94.39	95.25	97.57
Model price with $\lambda = 0$	114.8	117.6	118.2
Implied hazard λ (%/yr)	2.33	2.31	2.31
Pr(called at T_1)	51%	53%	60%
Lifetime Pr(conversion)	1.3%	1.5%	1.3%

Table 1: Last date, base case ($\kappa = 0.35$, $\eta = 100$bp).

Dispersion of the implied hazards across the three bonds, by spread volatility:

η (bp)	0	25	50	75	100	150	200
$D(\eta)$ (bp of hazard)	8.3	6.0	4.1	2.9	**2.3**	1.5	1.0

A stochastic refinancing spread is clearly needed (8.3bp with a deterministic one), but the dispersion keeps falling as η rises, so the cross-section does not pin η down; 100bp is kept as an assumption.

Over 66 weekly dates the mean dispersion is 6.0bp against 9.6bp for the deterministic rule. With Capital IQ benchmark proxies instead of the FRED curve the last-date dispersion is 2.7–7.6bp.

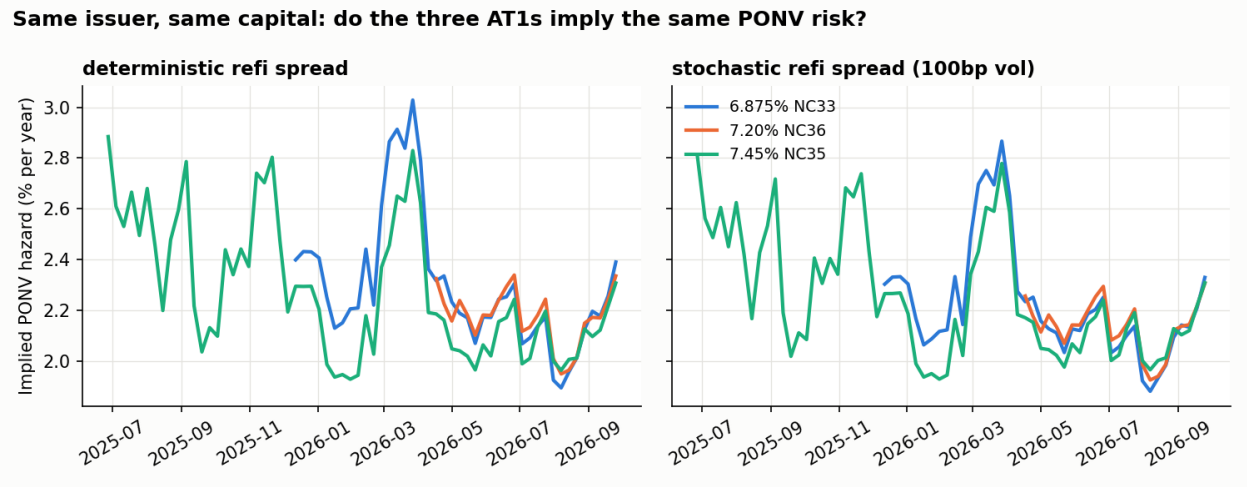
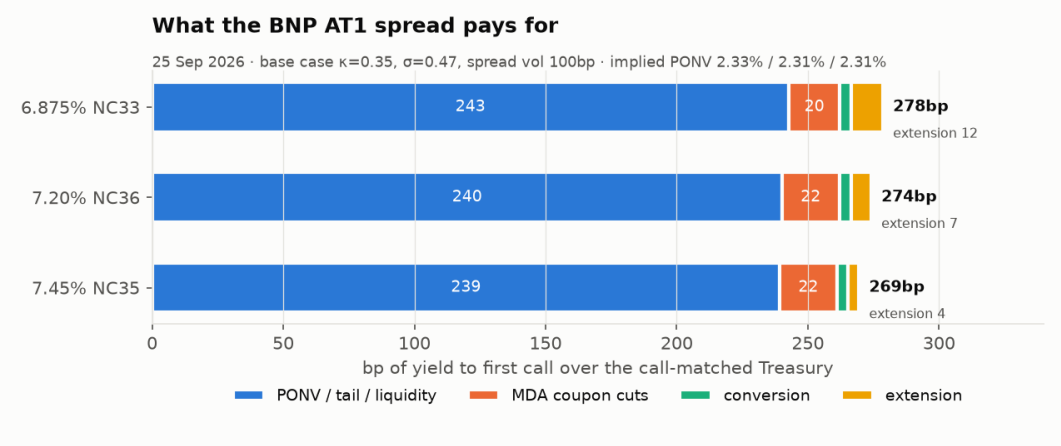


Figure 2: Implied PONV hazard per bond, weekly.

7.3 What the spread pays for

bp of yield-to-call spread	6.875% NC33	7.20% NC36	7.45% NC35
PONV / tail / liquidity	243	240	239
MDA coupon cuts	20	22	22
Conversion at the trigger	4	5	4
Extension	12	7	4
Total = market	279	274	269

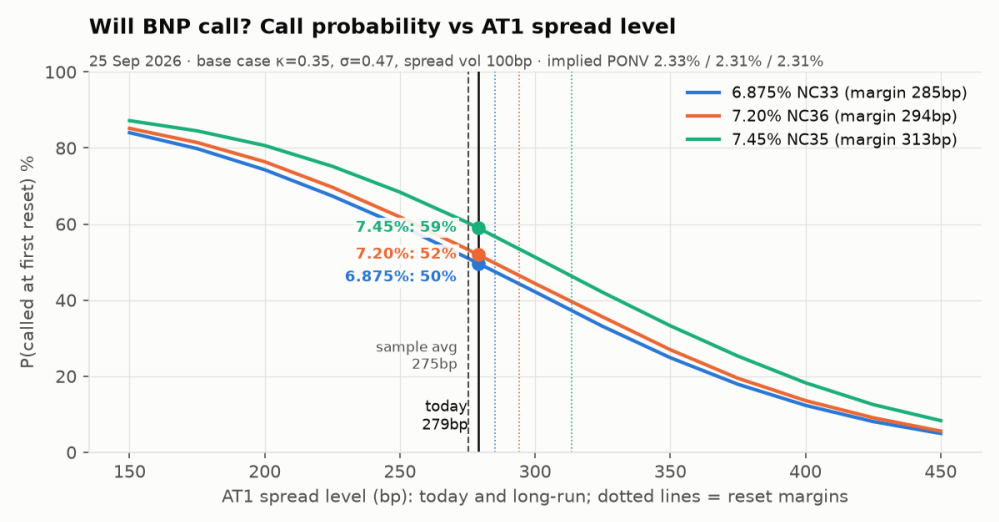


With BNP's capital dynamics, mechanical conversion is a 1.3–1.5% lifetime event worth ≈ 4bp. Almost all of the spread is compensation for regulatory write-down before the trigger, tail risk beyond the historical sample, and liquidity. The extension component decreases with the reset

margin, as the theory requires. The split between the premium and the MDA component depends on κ (§7.6); the total does not.

7.4 Call and extension

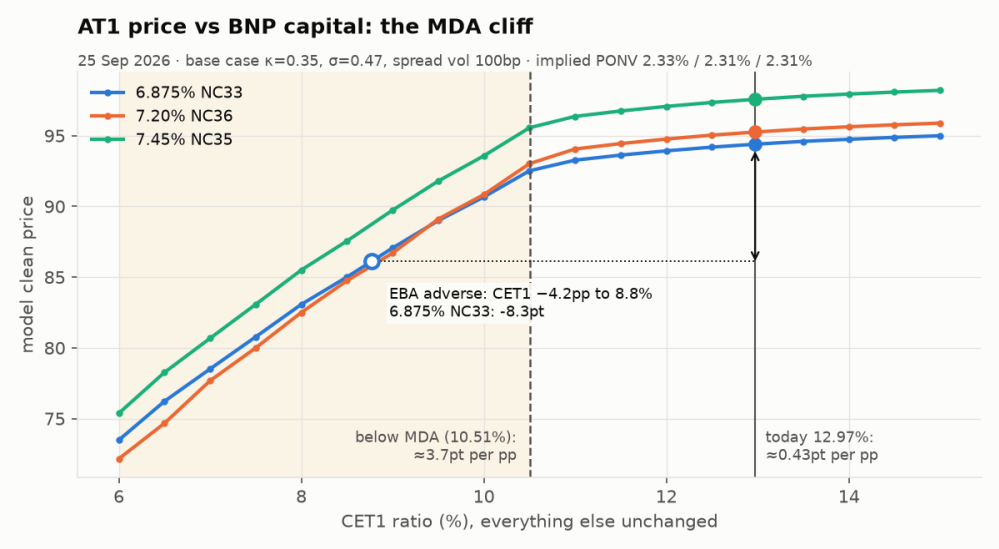
	6.875% NC33	7.20% NC36	7.45% NC35
Margin − today's AT1 spread (bp)	+6	+15	+34
Cost of the issuer's option (pts)	0.63	0.47	0.26
P&L, spreads +50bp (pts)	−3.4	−3.6	−3.3
of which from extension (pts)	−1.2	−0.9	−0.7



The low-margin bond is the most negatively convex. In the four weeks to 25 September 2026 the 10y Treasury rose 45bp; on top of that, the spread of the 6.875% widened by 34bp, against 10–12bp for the other two.

7.5 Stress and the capital cliff

Clean price change (points)	6.875% NC33	7.20% NC36	7.45% NC35
Rates +100bp	−5.2	−6.3	−6.1
CET1 −2pp	−1.2	−1.2	−1.2
CET1 −4.2pp (EBA adverse)	−8.3	−9.5	−8.9
AT1 spreads +200bp	−16.9	−16.8	−16.0
EBA adverse + spreads +200bp	−24.6	−25.6	−24.2
PONV hazard ×3	−26.8	−29.0	−27.9



The capital delta is ≈ 0.45 points per point of CET1 at 13% and ≈ 4 around and below the MDA threshold. Crossing $\bar{c}_{\text{MDA}}$ switches on three mechanisms at once: coupons are cut by (4), the call is blocked by (5), and the hazard rises through (3)–(7). For risk management, the distance that matters is the distance to the MDA threshold, not to the trigger.

7.6 How much can we trust the capital parameters?

Bootstrap. We simulate 1,000 CET1 histories on the observed quarterly dates with the fitted dynamics and re-estimate (κ, σ) on each. The 90% intervals are $\kappa \in [0.06,\ 0.48]$ and $\sigma \in [0.37,\ 0.52]$: volatility is well estimated, the speed of capital rebuilding is not. The half-life of a capital shock, $\ln 2/\kappa$, lies between 1.5 and 11 years (point estimate 4.2).

Average of the three bonds	$\kappa = 0.06$ (5%)	$\kappa = 0.165$ (history)	$\kappa = 0.35$ (base)	$\kappa = 0.48$ (95%)
Implied PONV (%/yr)	0.9	1.9	2.3	2.4
MDA coupon cost (pts)	10.9	4.4	1.7	1.2
Capital delta at the MDA threshold (pts/pp)	12.9	7.0	3.3	2.3
EBA adverse shock (pts)	−35.6	−18.7	−8.9	−6.4
Cross-bond dispersion D (bp)	8.6	4.0	2.3	1.9
History log-likelihood (relative to MLE)	−0.5	0	−1.5	−4.1

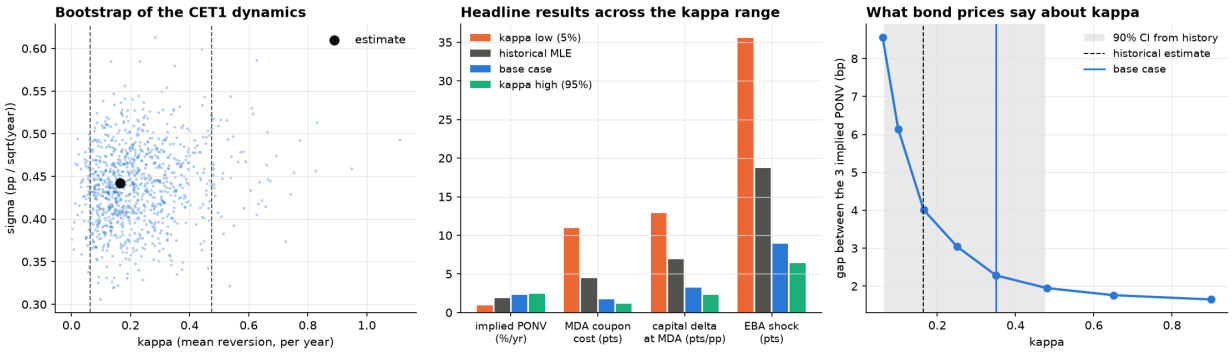


Figure 3: Left: bootstrap of (κ, σ). Centre: headline results across the κ range. Right: cross-bond dispersion of the implied hazard against κ.

What prices add. History and prices pull in opposite directions. The three bonds are mutually consistent only for fast capital rebuilding: D falls from 8.6bp at $\kappa = 0.06$ to 1.9bp at 0.48 and then flattens. The history, however, rejects values much above 0.35. The base case $\kappa = 0.35$ (half-life two years) lies inside the historical interval and close to the market plateau. It is also consistent with BNP's active capital management (a 13% target, managed RWA and distributions). The ordering of the bonds, the extension ranking and the steepening of the capital delta near the MDA threshold hold across the whole range. The level of the capital stress and the premium–MDA split do not, and are reported with this range.

Where the uncertainty sits. Each curve in Figure 4 is re-calibrated to today's price of the 6.875%, so all four scenarios agree at today's 13% CET1 (94.4). They separate only as capital falls: at the MDA threshold the price ranges from 80 ($\kappa = 0.06$) to 93 ($\kappa = 0.48$), and at 6% CET1 from 36 to 79. Prices identify the level of risk today, not the shape of the cliff.

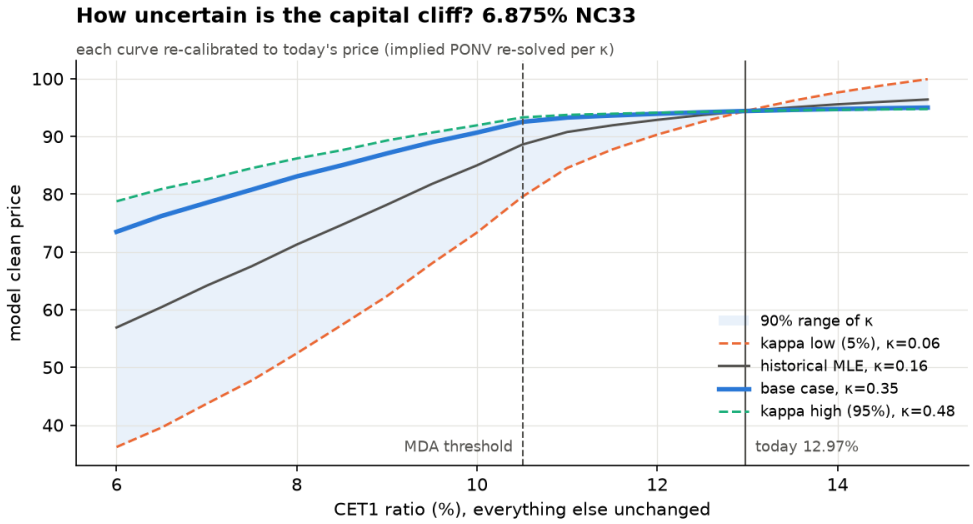


Figure 4: Price of the 6.875% NC33 against CET1 for four values of κ, each re-calibrated to today's price. Shaded: 90% bootstrap range of κ.

8 Limitations and extensions

1. **Measure.** CET1 is not traded. Its dynamics are estimated under the physical measure, and all remaining risk premia are absorbed by λ, which is therefore a risk-neutral hazard, not a default probability.
2. **Capital mean reversion.** κ is weakly identified by 50 quarters of history and drives the capital stress results (§7.6); the base case combines history and prices.
3. **Rates.** Today's FRED curve is used for all future resets; a short-rate model (Hull–White) would add rate-driven extension.
4. **Spread volatility.** η is not pinned down by the cross-section.
5. **Capital premium.** β controls the size of the capital cliff and is assumed. A cross-bank regression of AT1 spreads on MDA distance would calibrate it.
6. **Data.** Prices are evaluated rather than traded, and cover 15 months of tight spreads.
7. **Conversion.** A single stressed share price replaces a joint equity–capital process, and PONV recovery is set to zero.

A Parameters

Symbol	Meaning	Value	Source
κ, σ	CET1 mean reversion, volatility	0.35, 0.47pp	MLE 0.165/0.44; base (§7.6)
θ	CET1 target	13%	BNP strategic plan
λ_J, m_J, σ_J	stress jumps	0.12, 2.2pp, 0.5	EBA 2025 check
b, w	MDA floor, combined buffer	5.64%, 4.87%	requirement stack
c_{trig}	trigger	5.125%	term sheets
$a, \bar{s}, \eta, \rho$	spread OU	0.5, 275bp, 100bp, -0.3	sample, assumption
β	capital premium	1% per pp	assumption
ξ	share at trigger / today	0.2	assumption
λ	PONV hazard today	$\approx 2.3\%$	implied

B Code map

File	Content
`processes.py`	CET1 jump-diffusion (1), trigger times
`instrument.py`	MDA schedule (4), stylised term sheet
`pricer.py`, `risk.py`	original engine, Greeks, loss distribution
`market.py`	dated bonds, (2)–(7), pricing (9), implied λ
`calibrate_bnp.py`	calibration (§5)
`spread_and_stress.py`	decomposition, call analysis, stress
`parameter_uncertainty.py`	bootstrap, κ sensitivity (§7.6)